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Contact allergy in children with atopic dermatitis: a systematic review*

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Summary

The importance of contact allergy in children with atopic dermatitis is frequently debated. Previously, patients with atopic dermatitis were believed to have a reduced ability to produce a type IV immunological response. However, this belief has been challenged and authors have highlighted the risk of underestimating and overlooking allergic contact dermatitis in children with atopic dermatitis. Several studies have been published aiming to shed light on this important question but results are contradictory. To provide an overview of the existing knowledge, we systematically reviewed studies that report frequencies of positive patch test reactions in children with atopic dermatitis. We identified 436 manuscripts of which 31 met the inclusion criteria. Although the literature is conflicting, it is evident that contact allergy is a common problem in children with atopic dermatitis.

What's already known about this topic?

- The impaired skin barrier in atopic dermatitis facilitates the penetration of potential allergens.
- Children with atopic dermatitis are exposed to potential allergens in topical agents and emollients from an early age.

What does this study add?

- Contact allergy is a significant problem in children with atopic dermatitis and should always be considered in cases of recalcitrant atopic dermatitis.
- Children with atopic dermatitis may have unacknowledged contact allergies contributing to their skin symptoms.
- Children with atopic dermatitis seem to be at greater risk of sensitization to certain allergens, especially components of skincare products.

Atopic dermatitis (AD) is the most common inflammatory skin disease in childhood, affecting 15–30% of children.[1,2] Although the pathogenesis of AD is complex and multifactorial, the disease has a common phenotypic expression, characterized by dry and itchy skin with chronic or recurrent episodes of dermatitis at typical anatomical sites.[3] AD was traditionally considered an immune-mediated condition, driven primarily by T helper (Th) 2 cells,[4] and individuals with AD were considered less likely to have allergic contact dermatitis (ACD) owing to suppressed Th1-mediated cellular immunity.[5,6] Lately, this hypothesis has been challenged, and it is currently a topic of debate as to whether AD is caused primarily by a dysfunctional innate and adaptive immune system and an imbalance between Th1 and Th2 cells, or a primary skin barrier defect with immunological changes secondary to this, promoted by the invasion of trigger factors such as allergens and microbes.[7]

ACD is a type IV delayed hypersensitivity reaction that typically develops after repeated or prolonged topical exposure to chemical allergens. Clinically, AD and ACD may be difficult to distinguish as they both present as dermatitis and may

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coexist.[8] The relationship between the two is complex and the prevalence of contact allergy among children with AD is unknown. Whether children with AD have an altered risk of contact allergy compared with children without AD remains controversial, and studies have been conflicting. Theoretically, the impaired skin barrier in AD facilitates the penetration of potential allergens.[7] Indeed, children with AD are exposed to topical agents and emollients from an early age,[9] and the prolonged use of these agents could increase the risk of contact sensitization to both ingredients and vehicles. Several authors have highlighted the risk of underestimating and overlooking ACD in patients with AD.[8,10–12] However, experimental studies have found reduced sensitization among patients with AD compared with controls.[13–15]

The question remains whether patch testing to a greater extent should be used as a screening tool in these children, in order to exclude hidden allergies possibly maintaining or aggravating their skin symptoms. This article aims to review and summarize the existing knowledge on contact allergy and ACD among children with AD.

Methods

A literature search was performed through PubMed and MEDLINE on 6–7 February 2017 using the search items ('allergic contact dermatitis' OR 'contact allergy' OR 'contact sensitization' OR 'patch testing') AND ('children'). This was followed by a review of all titles and abstracts. English-language literature reporting the prevalence of sensitization to contact allergens in children and adolescents and providing exact information on the results of patch testing in individuals with AD was included. The search process is outlined in Figure 1.

Data extraction

The following data points were extracted: total number of children patch tested, total number of children with and without AD in the patch tested population, total number of children with and without AD who had at least one positive patch test reaction, and the frequency of positive patch test reactions to individual allergens. Furthermore, the criteria for the diagnosis of AD, the age span of the patch tested population, the patch test methodology, and the allergen series used for patch testing was noted for each study.

Statistics

Statistical analysis was performed with the statistical software program Stata 12·1 (StataCorp, College Station, TX, U.S.A.). The weighted average of prevalence rates was calculated as:

$$[(n_1 \times \%_1) + (n_2 \times \%_2) + (n_n \times \%_n)] \,/\, (n_1 + n_2 + n_n)$$

Comparison of proportions was made using the χ^2-test. Odds ratios with 95% confidence intervals of contact allergy in children with AD compared with children without AD were calculated, and a forest plot was constructed for the comparison of the results. Heterogeneity between studies was assessed using the I^2 statistic. Values of 25%, 50% and 75% were considered as indicating low, moderate and high heterogeneity, respectively.[16] Statistical significance was predefined as a P-value < 0·05.

Results

Children with and without atopic dermatitis referred for patch testing

Most studies addressing the issue of contact sensitization in children with AD report the prevalence of sensitization among children referred for patch testing. The authors then compare patch test results in children with AD with those in children without AD. Twenty-one out of 68 studies with this design provided exact information on the patch test results among children with and without AD (Table 1, Fig. 2). Overall, 41·7% of the children with AD were sensitized to at least one allergen, compared with 46·6% of children without AD.

Unselected children with atopic dermatitis

In eight studies, the authors patch tested unselected children with AD exclusively. In seven of these, a broad screening panel of allergens was used (Table 2). The weighted average sensitization rate was 26·2%. Angelini and Meneghini[17] used a series of allergens that differ substantially from those used in other studies with the same design, and the sensitization rate they reported is notably low. When excluding this study, the average sensitization rate was 34·8%.

In one study, the patch test results in children with AD were compared with those of healthy controls. Contact allergy was found in 45·2% of patients with AD, which was a significantly larger share than was found among controls (13·9%) (P < 0·01).[18]

Unselected children from the general population

Mortz *et al.*[19] patch tested 1146 unselected 8th grade schoolchildren (age 12–16 years) with TRUE Test panels 1 and 2. Overall, 15·2% had at least one positive patch test reaction. There was no significant association between contact allergy and past or present AD. In a recent study, 218 asymptomatic children aged 6–16 years were patch tested with the European Baseline Series. Twenty seven per cent (n = 59) of the children had a history of AD. Of these, 36% (n = 21) had at least one positive patch test reaction, compared with 28·9% (n = 46) in the group of children without AD (n = 159).[20]

Severity of atopic dermatitis and risk of contact sensitization

In three studies, the relationship between severity of AD evaluated by the SCORAD (Scoring Atopic Dermatitis) index[21] and risk of sensitization was assessed. Mailhol *et al.*[22] found that the risk of sensitization was significantly associated with the

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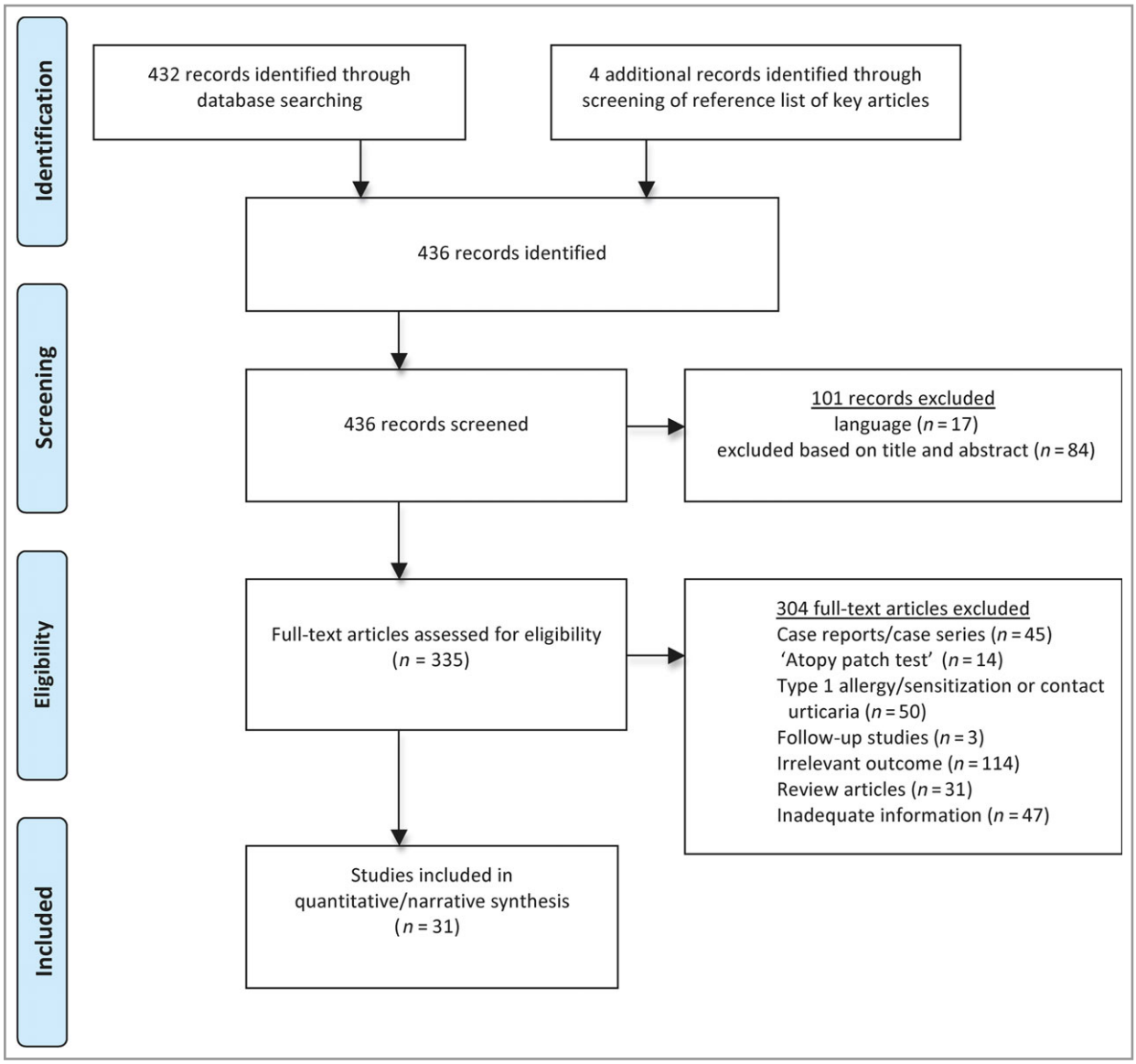

Fig 1. PRISMA flow diagram depicting the literature search and study selection.

severity of AD. Giordano-Labadie *et al.*[23] reported the opposite. Although the correlation was not significant, the authors found the lowest frequency of contact sensitization among children with severe AD (13·9%), compared with children with moderate (55·6%) and mild (30·6%) AD. Akan *et al.*[24] found no association between the severity of AD and the overall risk of contact sensitization.

Common allergens in children with atopic dermatitis

Adequate information on frequencies of positive patch test reactions was available in 18 studies. Tables 3 and 4 show the frequencies of positive patch test reactions to selected allergens. In studies where the two groups were compared, nickel sulfate yielded a positive patch test reaction in 17·2% (n = 704) of children with AD vs. 18·6% (n = 773) of children without AD (P = 0·10). In the studies on patch testing in children with AD exclusively, the weighted average of nickel contact allergy was 9·8% (n = 79).

Cobalt chloride and potassium dichromate were also frequent sensitizers,[10,12,18,25,26] as were fragrance markers (fragrance mix I, fragrance mix II, *Myroxylon pereirae* and hydroxyisohexyl-3-cyclohexenecarboxaldehyde).[11,18,23,27–30] When children with and without AD referred for patch testing were compared, the risks of sensitization to fragrance mix I and *Myroxylon pereirae* were significantly associated with AD (4·1% vs. 2·6%, P < 0·01; and 3·5% vs. 2·5%, P < 0·02, respectively). In studies on children with AD exclusively, the weighted average rate of sensitization to fragrance mix I was 2·4% (n = 13) and to *Myroxylon pereirae* 1·2% (n = 10).

In studies on children with and without AD, the average Amerchol L101 (lanolin alcohols) sensitization rate was

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Table 1 Studies reporting the prevalence of contact sensitization (positive patch test, PPT) in children and adolescents with and without atopic dermatitis referred for patch testing

Study	Country	Allergens	Children (n)	Age (years)	With AD			Without AD			P-value
					N	PPT, n (%)	95% CI of %	N	PPT, n (%)	95% CI of %	
Pambor 1992[25]	Germany	'11 most frequent allergens' + additional allergens as indicated	366	2–14	214	53 (24·8)	19·1–31·1	152	28 (18·4)	12·6–25·5	0·15
Wilkowska 1996[70]	Poland	29 allergens: two standard sets of allergens + 'medicinal set'	100	5–15	48	25 (52)	37·2–66·7	52	24 (46)	32·2–60·5	0·56
Shah 1997[71]	U.K.	European Standard Series, selected preservatives, vehicles and medicaments	83	6–16	29	11 (38)	20·7–57·7	54	30 (56)	41·4–69·1	0·13
Guillet 1997[27]	France	'Standard battery of allergens recommended by the ICDRG guidelines' + personal hygiene and cosmetic products[a]	152	3 months to 18 years	54	30 (56)	41·4–69·1	98	74 (76)	65·8–83·6	0·01
Manzini 1998[72]	Italy	European Baseline Series[b]	670	6 months to 12 years	506	209 (41·3)	37·0–45·7	164	73 (44·5)	36·8–52·5	0·47
Roul 1999[73]	France	European Baseline Series + additional allergens as indicated[c]	337	1–15	257	165 (64·2)	58·0–70·1	80	61 (76)	65·4–85·1	0·04
Fernandez Vozmediano 2005[74]	Spain	GEIDAC baseline series + additional allergens as indicated	96	< 15	27	14 (52)	31·9–71·3	69	38 (55)	42·6–67·1	0·78
Seidenari 2005[53]	Italy	Paediatric series of 30 of 46 allergens[d]	1094	7 months to 12 years	404	223 (55·2)	50·2–60·1	690	347 (50·3)	46·5–54·1	0·12
Beattie 2007[29]	U.K.	European Baseline Series. BCDG standard series[e]	114	3–15	27	10 (37)	19·4–57·6	83	51 (61)	50·1–71·9	0·03
Hogeling 2008[75]	Canada	NACDG standard series of 65 allergens + the patient's own products if indicated	100	4–18	41	30 (73)	57·1–85·8	59	40 (68)	54·4–79·4	0·56
de Waard-van der Spek 2009[54]	the Netherlands	TRUE Test panels 1 + 2 + tixocortol-17-pivalate, budesonide, three commonly used emollients + additional allergens as indicated	79	1–18	47	22 (47)	32·1–61·9	32	18 (56)	37·7–73·6	0·41
Milingou 2010[76, f]	Greece	European Baseline Series + additional allergens as indicated	232	≤ 16	73	32 (44)	32·2–55·9	159	79 (49·7)	41·7–57·7	0·40
Milingou 2010[76, g]	Greece	European Baseline Series + additional allergens as indicated	255	≤ 16	122	67 (54·9)	45·7–63·9	133	86 (64·7)	55·9–72·7	0·11
Bellino Fortina 2011[26]	Italy	Standard paediatric series of 30 allergens	321	≤ 3	137	84 (61·3)	52·6–69·5	184	116 (63·0)	55·6–70·0	0·76
Herro 2011[11]	U.S.A.	TRUE Test panels 1–3	101	6–18	54	48 (89)	77·4–95·8	47	31 (66)	50·7–79·1	< 0·01
Toledo 2011[77]	Spain	GEIDAC baseline series	111[h]	< 16	58	27 (47)	33·3–60·1	49	22 (45)	30·7–59·8	0·86
Kuljanac 2011[78]	Croatia	The Croatian standard series of 22 contact allergens	412	4–18	179	64 (35·8)	28·7–43·2	233	85 (36·5)	30·3–43·0	0·88
Schena 2012[10]	Italy	Standard series of 30 allergens	349	< 15	123	68 (55·3)	46·1–64·3	226	174 (77·0)	70·9–82·3	< 0·01
Belloni Fortina 2015[32]	Multinational	European Baseline Series	6708[i]	1–16	2644	997 (37·7)	31·9–45·5	2498	991 (39·7)	37·7–41·6	0·14

(continued)

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Table 1 (continued)

Study	Country	Allergens	Children (n)	Age (years)	With AD			Without AD			P-value
					N	PPT, n (%)	95% CI of %	N	PPT, n (%)	95% CI of %	
Belloni Fortina 2016[79]	Italy	Paediatric series of 30 allergens	2614	1–10	1283	600 (46·8)	44·0–49·5	1331	631 (47·4)	44·7–50·1	0·54
Smith 2016[80]	U.K.	BSCA baseline series + additional allergens as indicated	500	≤ 17	335	76 (22·7)	18·3–27·6	165	57 (34·5)	27·3–42·3	< 0·01
Lubbes 2017[28]	the Netherlands	European Baseline Series + additional allergens as indicated	1012[j]	< 18	526	252 (47·9)	43·6–52·3	395	185 (46·8)	41·8–51·9	0·74
Total			15 806		7188	3000 (41·7)	40·6–42·9	6953	3241 (46·6)	45·4–47·8	< 0·01

BCDG, British Contact Dermatitis Group; BSCA, British Society for Cutaneous Allergy; GEIDAC, Spanish Contact Dermatitis and Skin Allergy Research Group; ICDRG, International Contact Dermatitis Research Group; NACDG, North American Contact Dermatitis Group. [a]Children aged 3 months to 3 years were patch tested with a limited series of nine allergens. [b]During 1993–4, 120 children were patch tested with eight additional substances as the European Baseline Series changed. [c]Children < 5 years were patch tested with a shortened standard series of 17 allergens. [d]Children > 10 years were patch tested with 46 allergens. [e]The first 66 children were patch tested with the European Baseline Series. The remaining 48 children were patch tested with the BCDG standard series. [f]Study period 1980–1993. [g]Study period 1994–2007. [h]Information concerning atopic dermatitis was unavailable in four children. [i]Information on past or present atopic dermatitis was missing in 1566 children. [j]Information on past or present atopic dermatitis was missing in 91 children.

significantly higher among children with AD (11·4%, n = 23 vs. 4·2%, n = 9; P < 0·01), whereas there was no difference when comparing the weighted average sensitization rate to lanolin (4·1%, n = 156, vs. 4·0%, n = 142; P = 0·37). In studies on children with AD only, the weighted lanolin sensitization rate was 2·0% (n = 9). Among these, only Isaksson *et al.*[12] reported the frequency of sensitization to Amerchol L101, and found that 11% of 82 children with AD were sensitized. All reactions were considered relevant. The risk of contact allergy was significantly associated with hand or foot eczema.

In studies comparing children with and without AD, few children with AD were sensitized to tixocortol pivalate (n = 23) or budesonide (n = 6), and the weighted average sensitization rates for the two markers did not differ significantly between the two groups (tixocortol pivalate 0·8%, n = 23 vs. 0·5%, n = 14; P = 0·16, budesonide 0·2%, n = 6 vs. 0·08%, n = 2; P = 0·24). In the eight studies on patients with AD only, three children were sensitized, all to tixocortol-21-pivalate.[22,24,31]

In two studies, the authors aimed to evaluate contact allergy to topical steroids in children with AD.[22,31] Mailhol *et al.*[22] patch tested 641 children and adolescents under the age of 16 years with seven components of common topical agents used in the treatment of AD. Patch test readings were performed at 48 and 72 h. A positive patch test to one or more of the seven substances was found in 6·2%, the most frequent sensitizers being the child's own emollient (3%) and chlorhexidine (2·7%). Only one of the 641 children reacted to one of two topical corticosteroid markers (tixocortol-21-pivalate). In the study by Foti *et al.*[31] one of the 71 patch tested children with AD was sensitized to the screening markers tixocortol-21-pivalate and hydrocortisone-17-butyrate. Patch test readings were performed on days 2, 4 and 7.

Three studies reported rates of sensitization to sesquiterpene lactone mix among children with AD vs. children without.[28,29,32] Although few children reacted to the allergen, the average sensitization rate was significantly higher among children with AD (1·6%, n = 25 vs. 0·2%, n = 2; P < 0·01). Two studies reported on rates of sensitization to compositae mix.[26,29] Overall two children reacted to compositae mix and both had AD. Four of the studies testing only children with AD reported on sesquiterpene lactone mix sensitization rates, and found a positive reaction in three of 434 tested children (0·7%).[12,18,23,24]

Discussion

Based on the 21 studies included in the present review that compared the results of patch tested children with and without AD, the prevalence of contact allergy was significantly higher among children without AD. However, the sensitization rates reported vary considerably, as do the studies regarding size, study design, allergens used for patch testing, the patch test methodology, age and sex distribution, and criteria for AD, which makes it difficult to draw any strong conclusions regarding the relationship of AD and contact allergy.

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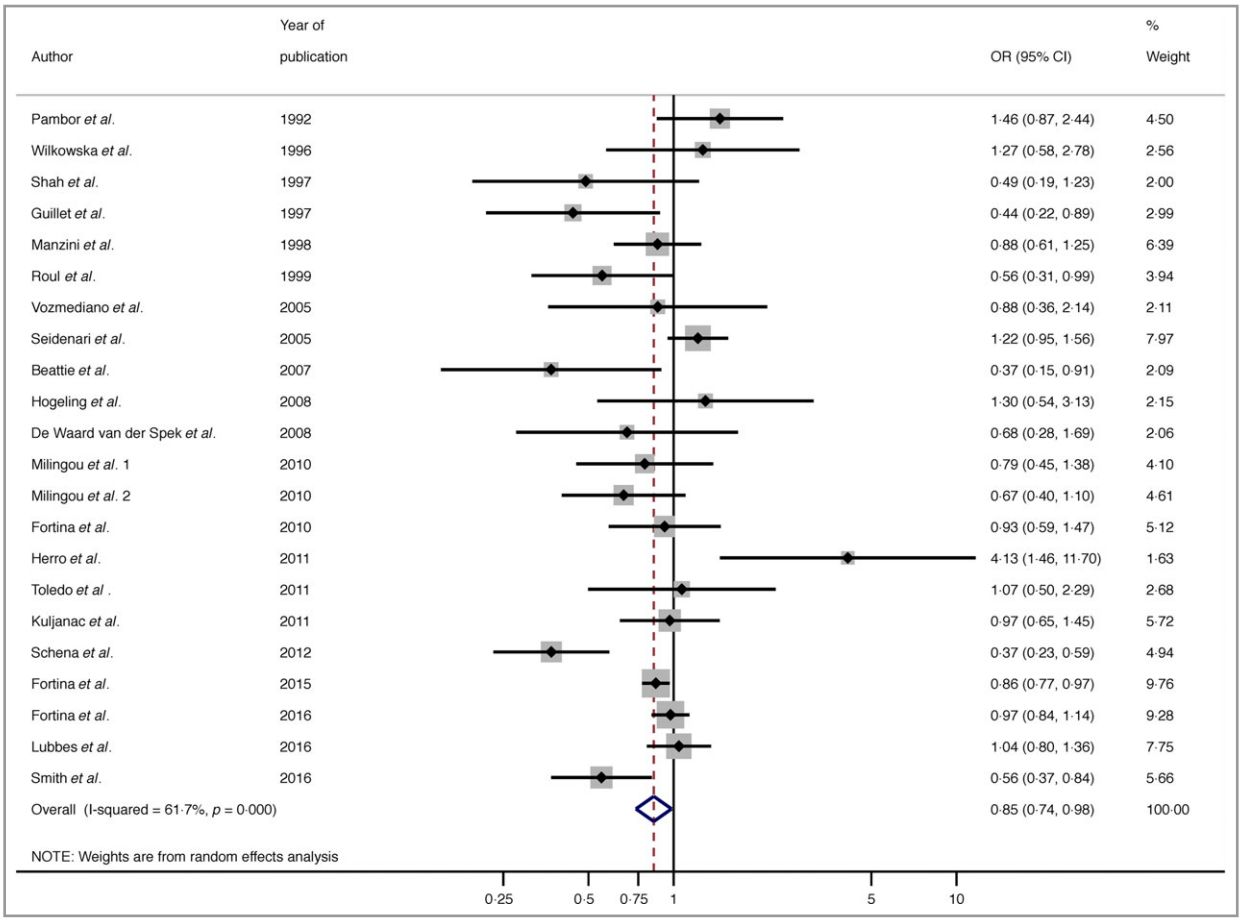

Fig 2. Forest plot depicting odds ratios (ORs) with confidence intervals (CIs) from the studies comparing sensitization rates in children with and without atopic dermatitis (see also Table 1).

This finding may support the previous hypothesis of individuals with AD being less likely to have ACD due to suppressed Th1-mediated cellular immunity.[5,13] However, it could also reflect a selection bias showing that children without AD are more carefully selected for patch testing (where the suspicion of ACD is high), whereas children with AD may be referred for patch testing because of aggravation of the AD.

The results of the studies that exclusively patch tested children with AD indicate that approximately one-third of children with AD may have one or more hidden contact allergies, possibly contributing to or maintaining the skin symptoms. Unfortunately only a few studies mention the severity of AD at the time of patch testing,[19,22–24] which would have been valuable information considering the results of the studies that originally found an altered sensitization threshold in patients with AD that was inversely correlated to the degree of severity.[13,14]

In the studies reviewed, metals were the most common allergens among patch tested children with AD and, as in patch tested children and adults worldwide,[32,33] nickel was the most common allergen. Children become sensitized to nickel at an early age[34] and the exposures are numerous.[35–38] Previous studies have suggested an increased risk of nickel sensitization in patients with AD.[39] The association between AD and filaggrin mutations is well established.[40] A positive association between filaggrin mutations and contact sensitization to nickel has previously been demonstrated among German adults.[41] This finding was later confirmed in Danish adults without ear piercings.[42] In the studies reviewed for this paper, we did not find an overall significant difference in nickel sensitization between children with and without AD. We speculate that this could reflect the role of ear piercing. According to the bypass theory, skin penetration events such as ear piercing bypass the skin barrier, thereby circumventing the nickel-chelating effect of skin filaggrin and other proteins that are downregulated in individuals with AD.[43]

Reactions to cobalt chloride and potassium dichromate were also frequent among the children with AD. Previous studies have reported the same pattern of sensitization in patients with AD.[44–46] In a study by Malajian and Belsito,[47] positive patch test reactions to nickel sulfate, cobalt chloride and potassium dichromate were significantly more frequent among patients with AD than in controls. The three metal allergens are known to occur concomitantly,[48] and the irritating

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Table 2 Studies reporting the prevalence of contact sensitization (positive patch test, PPT) in unselected children and adolescents with atopic dermatitis (AD)

Study	Country	Allergens	Diagnosis of AD	Children (n)	Age (years)	PPT, n (%)	95% CI of %	Top three allergens	Relevance (%)
Angelini 1977[17]	Italy	Standard series of 24 allergens	Not given	270	< 12	25 (9·3)	6·1–13·4	Potassium dichromate, neomycin sulfate, cobalt chloride	Not given
Oranje 1994[30]	The Netherlands	European Baseline Series	Rajka	33	5–15	14 (42)	25·5–60·8	Nickel sulfate, cobalt chloride, balsam of Peru	65·0
Giordano-Labadie 1999[23]	France	European Baseline Series, tixocortol pivalate, budesonide, applied emollient and additional allergens as indicated	Hanifin & Rajka	114	< 16	49 (43·0)	33·8–52·6	Nickel sulfate, fragrance mix, lanolin	100·0
Foti 2005[31]	Italy	Paediatric series of 20 allergens, tixocortol pivalate, budesonide and hydrocortisone-17-butyrate	Not given	71	5–12	10 (14)	7·0–24·4	Nickel sulfate, mercaptobenzothiazole, lanolin	Not given
Silny 2013[18]	Poland	Paediatric series of 12 allergens and European Baseline Series[a]	Hanifin & Rajka	104	1–20	47 (45·2)	35·4–55·3	Nickel sulfate, cobalt chloride, potassium dichromate	Not given
Akan 2015[24]	Turkey	TRUE Test	Hanifin & Rajka	134	≤ 17	45 (33·8)	25·7–42·2	Nickel sulfate, methylchloroisothiazolinone, thiomersal	49·0
Isaksson 2015[12]	Sweden	Children's Baseline Series (30 allergens) and corticosteroid series (24 allergens)	Not given	82	5–14	22 (27)	17·6–37·8	Amerchol L101, potassium dichromate, nickel sulfate	42·8
Total				808	1–20	212 (26·2)	23·2–29·4		

[a]Children aged 1–5 years were tested with an abbreviated paediatric series of 12 allergens.

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Table 3 Frequencies of positive patch test reactions to selected allergens in children with and without atopic dermatitis (AD)

	Nickel sulfate		Potassium dichromate		Fragrance mix I		*Myroxylon pereirae*		Lanolin		Amerchol L101		Tixocortol-21-pivalate		Budesonide	
Study	AD	No AD	AD	No AD	AD	No AD	AD	No AD	AD	No AD	AD	No AD	AD	No AD	AD	No AD
Pambor[25]	39/214	20/142	7/214	2/214	–	–	–	–	–	–	–	–	–	–	–	–
Wilkowska[70]	3/48	4/52	13/48	8/52	–	–	–	–	2/48	1/52	–	–	–	–	–	–
Guillet[27]	12/54	34/98	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Seidenari[53]	62/404	57/690	20/404	22/690	14/404	16/690	10/404	13/690	46/404	65/347	–	–	–	–	–	–
Beattie[29]	3/27	19/83	0/27	2/83	3/27	5/83	0/27	3/83	0/27	5/83	1/27	2/83	–	–	–	–
de Waard-van der Spek[54]	13/47	4/32	3/47	2/32	1/47	1/32	–	–	2/47	0/32	–	–	0/47	0/32	0/47	0/32
Bellino Fortina[26]	38/137	48/184	15/137	14/184	0/137	0/184	3/137	1/184	0/137	3/184	–	–	1/137	2/184	1/137	0/184
Herro[11]	19/54	12/47	6/54	2/47	10/54	0/47	11/54	1/47	13/54	5/47	–	–	6/54	1/47	–	–
Bellino Fortina[32]	434/2591	506/2444	174/2556	130/2414	63/2577	46/2424	68/2278	59/2035	52/2539	46/2408	–	–	9/2255	10/1988	3/2238	2/1971
Lubbes[28]	81/514	69/394	16/514	15/395	63/521	32/395	29/518	8/405	41/524	17/395	22/174	7/132	7/519	1/394	2/519	0/394
Total	704/4090	773/4166	254/4001	197/4111	154/3767	100/3855	121/3418	85/3444	156/3780	142/3548	23/201	9/215	23/3012	14/2645	6/2941	2/2581
Weighted average % (95% CI)	17·2 (16·1–18·4)	18·6 (17·4–19·8)	6·3 (5·6–7·1)	4·8 (4·2–5·5)	4·1 (3·5–4·8)	2·6 (2·1–3·1)	3·5 (2·9–4·2)	2·5 (2·0–3·0)	4·1 (3·5–4·8)	4·0 (3·4–4·7)	11·4 (7·4–16·7)	4·2 (1·9–7·8)	0·8 (0·5–1·1)	0·5 (0·3–0·9)	0·2 (0·07–0·4)	0·08 (0·01–0·3)

CI, confidence interval.

Table 4 Frequencies of positive patch test reactions to selected allergens in children with atopic dermatitis

Study	Nickel sulfate	Potassium dichromate	Fragrance mix I	*Myroxylon pereirae*	Lanolin	Amerchol L101	Tixocortol-21-pivalate	Budesonide
Angelini[17]	2/270	9/270	–	1/270	–	–	–	–
Oranje[30]	9/33	0/33	1/33	2/33	1/33	–	–	–
Giordano-Labadie[23]	17/114	3/114	5/114	3/114	5/114	–	–	–
Foti[31]	4/71	1/71	0/71	0/71	2/71	–	1/71	0/71
Mailhol[22]	–	–	–	–	–	–	1/641	0/641
Silny[18]	26/104	22/104	3/104	2/104	0/104	–	0/104	0/104
Isaksson[12]	4/82	6/82	2/82	1/82	–	9/82	0/82	0/82
Akan[24]	17/134	3/134	2/134	1/134	1/134	–	1/134	0/134
Total	79/808	44/808	13/538	10/808	9/456	9/82	3/846	0/391
Weighted average % (95% CI)	9·8 (7·8–12·0)	5·4 (4·0–7·2)	2·4 (1·3–4·1)	1·2 (0·6–2·2)	2·0 (0·9–3·7)	11·0 (5·1–19·8)	0·4 (0·07–1·0)	–

CI, confidence interval.

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properties of potassium dichromate could in theory cause a false-positive reaction in AD skin.[46] However, especially reactions to potassium dichromate were relevant in a substantial proportion of the reviewed cases.[12,23,24] As chromium is rarely released from jewellery,[42] the exposure to chromium is primarily percutaneous and the bypass theory therefore does not apply. Thus, our finding may indicate an increased risk of metal allergy in patients with AD.

Fragrance is a common sensitizer in children.[49] Several studies have previously found significantly higher prevalences of fragrance contact allergy among adult patients with AD compared with patients without AD.[11,39,44,50] From the current review we found that fragrance was a common sensitizer in children with AD who were patch tested because of deterioration of the AD or because of suspected ACD. However, fragrance allergy does not seem to be as common among children with AD without aggravating skin symptoms, namely children who were screened for concurrent contact allergy by patch testing. We speculate that this could reflect that children with AD, implying sensitive skin since early childhood, to a large extent are treated with fragrance-free emollients and skincare products carefully selected by the caregiver. However, in agreement with previous data, these children do seem to be at greater risk of sensitization to fragrance allergens.

In several of the reviewed studies, a significant proportion of children with AD reacted to Amerchol L101 and/or lanolin.[11,12,51–54] Amerchol L101 is a lanolin derivative that is used as an emollient and emulsifier in ointments and in cosmetic products such as soaps, shampoos, hair conditioners, aftershaves, baby and bath oils, hand sanitizers, lipsticks and creams.[29] Six of the reviewed studies patch tested with Amerchol L101 and all reported relatively high sensitization rates.[12,23,28,29,51,52] It has previously been advised to patch test with both lanolin alcohol and Amerchol L101 to increase the chances of detecting lanolin allergy.[55] From the reviewed literature, it seems to be a frequent cause of contact allergy and ACD in children with AD. However, Amerchol L101 has a mineral oil component that constitutes 5% of the test substance, and it is therefore possible that positive reactions to this allergen to some extent could be irritant reactions falsely interpreted as positive reactions.[12]

Although corticosteroids are not potent allergens,[56] contact sensitization and ACD from topical corticosteroids is not infrequent.[57] Since early childhood, children with AD are exposed to various topical treatments including topical corticosteroids. The combination of a dysfunctional skin barrier and prolonged exposure should in theory increase the risk of sensitization to ingredients of the treatment products. In previous studies an association between AD and contact allergy to topical corticosteroids was suggested.[56,58] It does not seem to constitute a significant problem among children with AD,[22–24,31,54] although delayed positive reactions may have been missed in the reviewed studies.[59] In the study by Mailhol *et al.*,[22] only one of 641 children reacted to one of the corticosteroid markers. However, patch test readings were performed only at day 2 and day 3. With a later reading (day 7), more allergic reactions to corticosteroids might have been detected. It is well known that diagnosing contact allergy to corticosteroids can be a major challenge, and the particular importance of late readings has frequently been stressed.[60,61] The possibility of a complicating contact allergy to topical treatments should not be ignored in the evaluation of a child with recalcitrant eczema.

Previous studies and case reports have suggested an increased risk of sensitization to plant allergens in patients with AD,[62–65] but the causality of this relationship is unclear. Although few of the reviewed studies provide information on contact allergy to sesquiterpene lactone mix and compositae mix, positive reactions were significantly associated with AD. Overall, compositae allergy is not considered common in children,[32,65] but patients with AD appear to be at greater risk of sensitization to compositae.[64] Like airborne allergens such as methylisothiazolinone, compositae contact dermatitis can mimic AD,[66,67] and the risk of misdiagnosing a child with AD who in fact has contact allergy to compositae exists. Thus, it is a relevant allergy to consider, especially in cases with dermatitis located on the hands or face and aggravation of skin symptoms during spring or summer.[64,65] Furthermore, these children should be warned against topical use of compositae-containing cosmetics and herbal remedies.[68]

One limitation of this review is the heterogeneity of studies that are compared. Data from studies using different populations and patch test methodologies were combined, which weakens the calculated weighted prevalence to some extent. Another limitation is that only one author reviewed the manuscripts and conducted data extraction for the search. This could hypothetically have biased the results. To provide transparency in the search methods and to conform to the PRISMA statement,[69] the search strategy and method of data extraction are described explicitly in Figure 1 and in the Methods section above.

In conclusion, although the literature is conflicting, it is evident that contact allergy is a common problem in children with AD. Children with AD seem to be at greater risk of sensitization to certain allergens including metals and components of skincare products. As emollients are a part of the basic treatment of AD, caregivers should be advised to use unscented products with as few components as possible, and preservatives with a low sensitizing potential. Patch testing may be relevant as a screening tool in the management of children with AD and should always be considered in cases of recalcitrant AD or whenever ACD is suggested by the child's history.

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